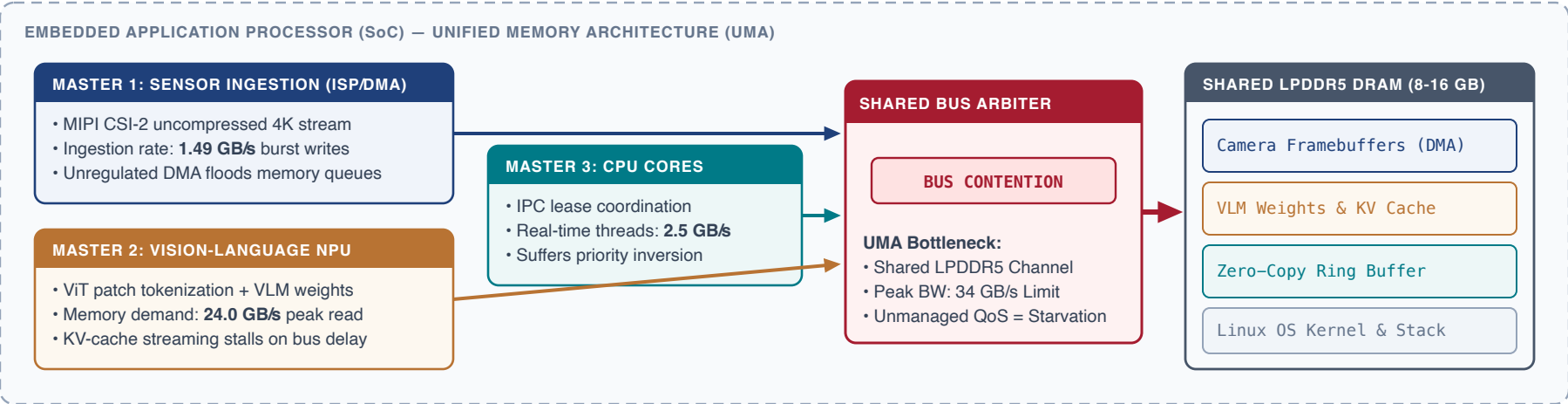


UMA MEMORY BUS CONTENTION & THE PRE-INFERENCE INGESTION TAX

Hardware bus collision between unthrottled camera DMA, NPU tensor streaming, and CPU real-time threads



NAIVE APPROACH: MEMCPY & UNMANAGED BUS BURSTS

Nominal Latency (P50):

22.4 ms

Tail Latency (P99):

135.8 ms

- **3x User/Kernel Space Copies:** V4L2 kernel buffer → OpenCV Mat → Tensor tensor.
- **Bus Starvation:** Camera bursts lock out NPU weight streaming during critical tokens.
- **Causal Consequence:** Breaches Freshness Wall ($\Delta t_{fresh} > 60\text{ ms}$), triggering motor aborts.

PHYSICAL AI CONTRACT: ZERO-COPY DMA + HARDWARE QoS

Nominal Latency (P50):

5.8 ms

Tail Latency (P99):

8.2 ms (Deterministic)

- **Zero-Copy DMABUF Pipeline:** ISP writes directly to memory-mapped tensor buffer.
- **Hardware QoS Slots:** Strict arbiter bandwidth caps guarantee real-time CPU/NPU access.
- **Bounded Metrology:** Bounded $P99 \leq 8.2\text{ ms}$ preserves metric state estimation freshness.